

Q1) Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?

Ans: A Bayesian Network shows the relationship between different machine conditions. We can use temperature, vibration, pressure, & oil level as inputs. Using past machine data, it learns the probability of failure. If the probability is high, maintenance ~~be~~ can be done before the machine fails.

Q3) If a Company wants to build a speech-recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?

Ans: A HMM is useful because speech is a sequence that changes with time. The audio features are observations, while phonemes or words are treated as hidden states. The model learns how one speech state changes to another. It finds the most probable sequence of sounds & converts it into words.

Q) Why do CRFs often outperform HMMs in NER? ~~Conditional Random Fields~~ ^{Fields}

Ans: CRFs can use many features such as words, capitalisation, prefixes, suffixes & surrounding words. They also consider the relationship between neighboring labels. HMMs make stronger assumptions & are less flexible. CRFs can use more context, so they often give better results in Named Entity Recognition (NER).

Q5) Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?

Ans: Conditional Independence means two variables become independent when another variable is known. For eg, fever & cough can be considered independent when the disease is known. This reduces the number of calculations required. Therefore, it makes probabilistic inference faster & simpler.